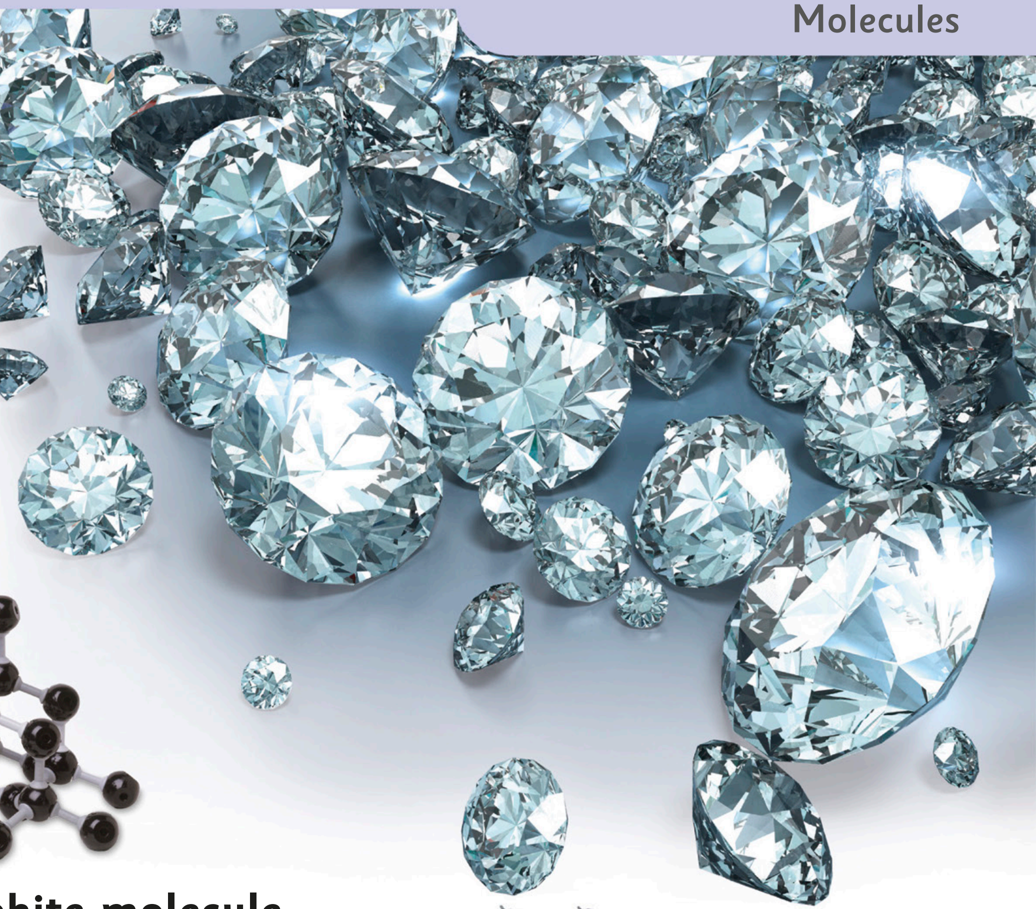


Diamond is made into jewels that are almost indestructible.

Diamond molecule

Diamond is the hardest natural substance known. Its hardness comes from the way the carbon atoms in diamond are arranged. Each atom is joined by strong bonds to four neighboring atoms.

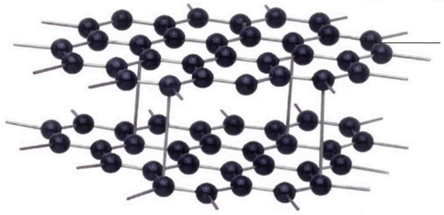
Each group of five atoms in diamond forms a pyramid shape. This shape makes diamond amazingly strong.

Turn and learn

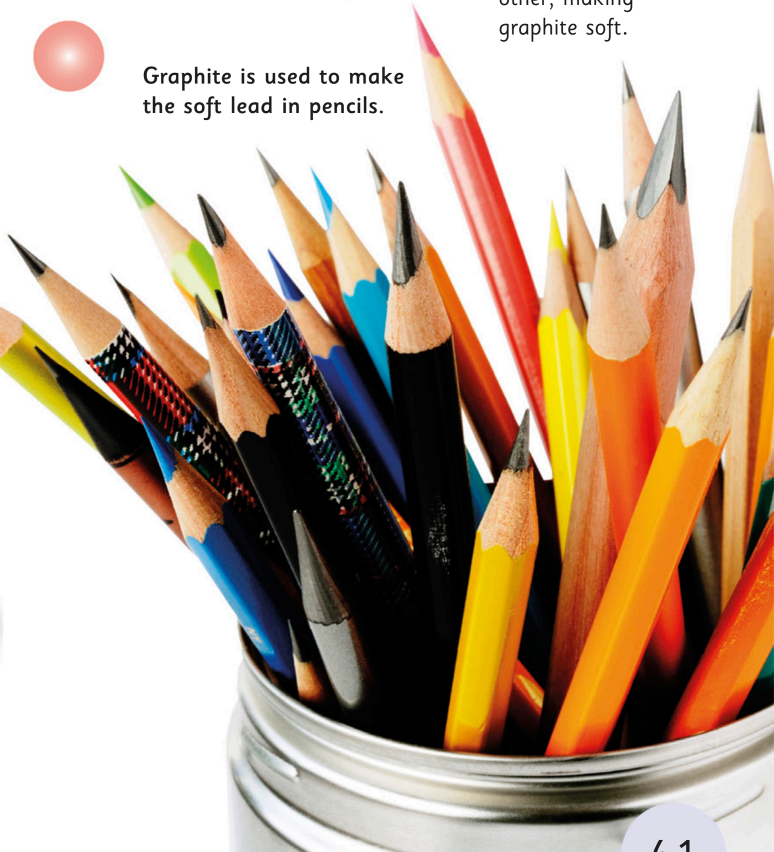
Changing states: **pp. 56–57**
Minerals: **pp. 104–105**

Graphite molecule

Graphite, like diamond, is also made of carbon atoms, but the atoms are arranged in a different way, making graphite very soft.

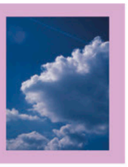


Each carbon atom in graphite is joined to only three neighbors. The atoms form layers that slip over each other, making graphite soft.



Graphite is used to make the soft lead in pencils.

Evaporate: As a liquid heats up, its molecules speed up until they move fast enough to float away as gas.



Condense: When gas molecules lose energy and slow down, they stick together and form liquid.

A gas can fill any container it's put in. If there's no lid to seal the container, the gas will escape into the air.